

Progress Test 3.C — Thermo-S1

3.C · Equations of state of real gases

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The co-volume of carbon dioxide is $b = 4.27 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$. Express it in $\text{cm}^3 \text{ mol}^{-1}$.

$1 \text{ m}^3 = (10^2 \text{ cm})^3 = 10^6 \text{ cm}^3$	$b = 4.27 \times 10^{-5} \times 10^6 \text{ cm}^3 \text{ mol}^{-1}$
$b = 42.7 \text{ cm}^3 \text{ mol}^{-1}$	

Marking

①
0 1

Marking scheme

① 1 pt $42.7 \text{ cm}^3 \text{ mol}^{-1}$.

Exercise 2. The Van der Waals model

(5 marks)

Write the Van der Waals equation of state for n moles. Explain the physical meaning of the term involving b and of the term involving a . What does the equation reduce to when $a \rightarrow 0$ and $b \rightarrow 0$?

$\left(p + a \frac{n^2}{V^2}\right) (V - nb) = nRT$ nb is the co-volume. Real molecules have a non-zero size, so the volume actually available to them is $V - nb$ (the excluded volume is about four times the molecules' own volume). This repulsive effect raises the pressure. $a n^2/V^2$ is the internal pressure. Molecules attract each other, so a molecule heading for the wall is held back by its neighbours and the measured pressure is lower. The term is proportional to $(n/V)^2$: the number of molecules reaching the wall and the number of neighbours pulling each one back both scale with n/V. When $a \rightarrow 0$ and $b \rightarrow 0$: $pV = nRT$, the ideal gas.

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts The equation, with n^2/V^2 and $V - nb$.
 ② 2 pts b : excluded volume or finite size (1 pt); a : attraction, internal pressure (1 pt).
 ③ 1 pt Limit: ideal gas.

Exercise 3. The Amagat diagram

(4 marks)

In the Amagat diagram, when $p \rightarrow 0$ the compressibility factor Z tends to:

- ☒ 1 ☐ 0 ☐ $+\infty$ ☐ -1

Why: at very low density every gas behaves as an ideal gas, for which $Z = 1$.

At moderate pressure, $Z < 1$ for most gases because:

- ☒ attractive interactions dominate ☐ the molecules' size dominates
☐ the gas is ideal ☐ the temperature is too high

Why: attractions reduce the pressure (or the volume) below the ideal value. At high pressure the excluded volume wins and $Z > 1$.

The Boyle temperature T_B is the temperature at which:

- ☒ $Z \approx 1$ over a wide range of p ☐ $Z = 0$
☐ the gas liquefies ☐ $a = 0$

Why: at T_B the attractive and repulsive corrections compensate over a wide range of pressures (Fig. 9).

According to Tab. 7, the gas with the strongest intermolecular attractions, and the highest critical temperature, is:

- ☐ He ☐ H₂ ☐ N₂ ☒ H₂O

Why: $a(\text{H}_2\text{O}) = 0.554 \text{ Pa m}^6 \text{ mol}^{-2}$ and $T_c = 647 \text{ K}$: a and T_c are correlated.

Exercise 4. Carbon dioxide under pressure

(4 marks)

One mole of CO₂ occupies the volume V at temperature T . Compute (a) the pressure p_{id} predicted by the ideal gas law; (b) the pressure p predicted by the Van der Waals equation; (c) the corresponding compressibility factor Z . Interpret.

$$n = 1.0 \text{ mol} \quad V = 0.50 \text{ L} \quad T = 300 \text{ K} \quad RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$$

$$a = 0.36 \text{ Pa m}^6 \text{ mol}^{-2} \quad b = 4.0 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1} \quad (\text{rounded values for CO}_2)$$

(a) $p_{\text{id}} = \frac{nRT}{V} = \frac{2.5 \times 10^3}{5.0 \times 10^{-4}} \quad p_{\text{id}} = 5.0 \times 10^6 \text{ Pa} \quad (50 \text{ bar})$

(b) $p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$, with

$$V - nb = 5.0 \times 10^{-4} - 0.40 \times 10^{-4} = 4.6 \times 10^{-4} \text{ m}^3:$$

$$\frac{nRT}{V - nb} = \frac{2.5 \times 10^3}{4.6 \times 10^{-4}} \approx 5.4 \times 10^6 \text{ Pa}$$

$$a \frac{n^2}{V^2} = \frac{0.36}{(5.0 \times 10^{-4})^2} = 1.44 \times 10^6 \text{ Pa}$$

$$p \approx 4.0 \times 10^6 \text{ Pa} \quad (40 \text{ bar})$$

(c) $Z = \frac{pV}{nRT} = \frac{p}{p_{\text{id}}} \quad Z \approx 0.80 < 1$. Attraction dominates, as on the Amagat diagram at moderate pressure.

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	

Marking scheme

- ① 1 pt $p_{\text{id}} = 5.0 \times 10^6 \text{ Pa}$.
- ② 1 pt Van der Waals equation solved for p (literal).
- ③ 1 pt $p \approx 4.0 \times 10^6 \text{ Pa}$.
- ④ 1 pt $Z \approx 0.8$, attractions dominate.

Examiner's tip

Convert first: $0.50 \text{ L} = 5.0 \times 10^{-4} \text{ m}^3$. Then $(5.0 \times 10^{-4})^2 = 25 \times 10^{-8} = 2.5 \times 10^{-7} \text{ m}^6$: square the mantissa *and* the power of ten.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20